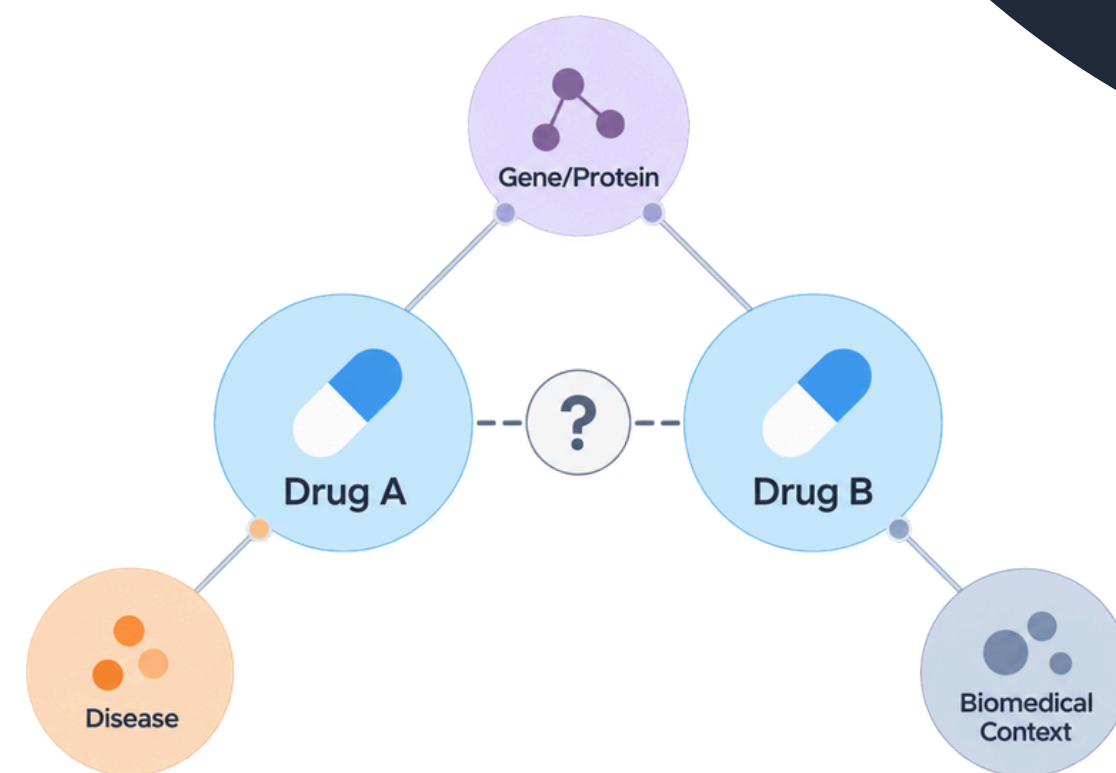


What Biomedical Information Helps Predict Drug Interactions?

Effect of Biomedical Knowledge Graph Composition on R-GCN-Based Drug–Drug Interaction Prediction

Team CHEERS
Pusan National University

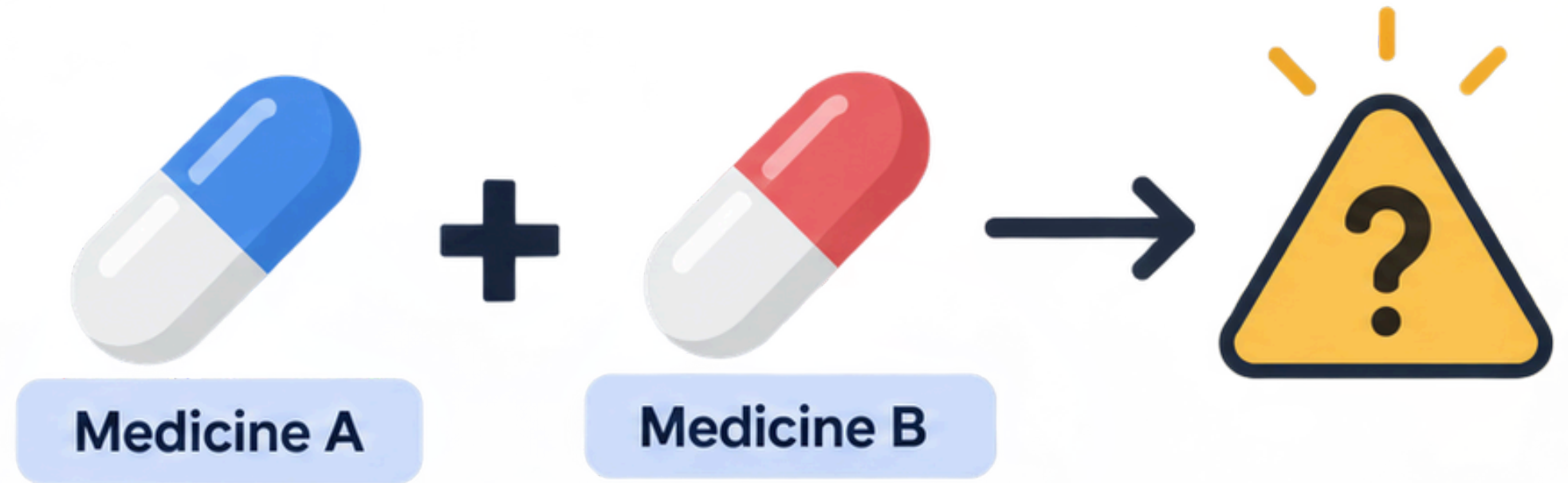
BYAMBASUREN TUVSHINJARGAL
GALBADRAKH BUYANDELGER
BAVUUJAV DELGERBAYAR



Why Do Drug Interactions Matter?

Medicines help us treat illnesses and feel better.

But taking two or more medicines together can sometimes cause unexpected interactions.



Drug-Drug Interaction (DDI)

When one medicine affects how another medicine works.



There are too many possible drug combinations to study one by one.

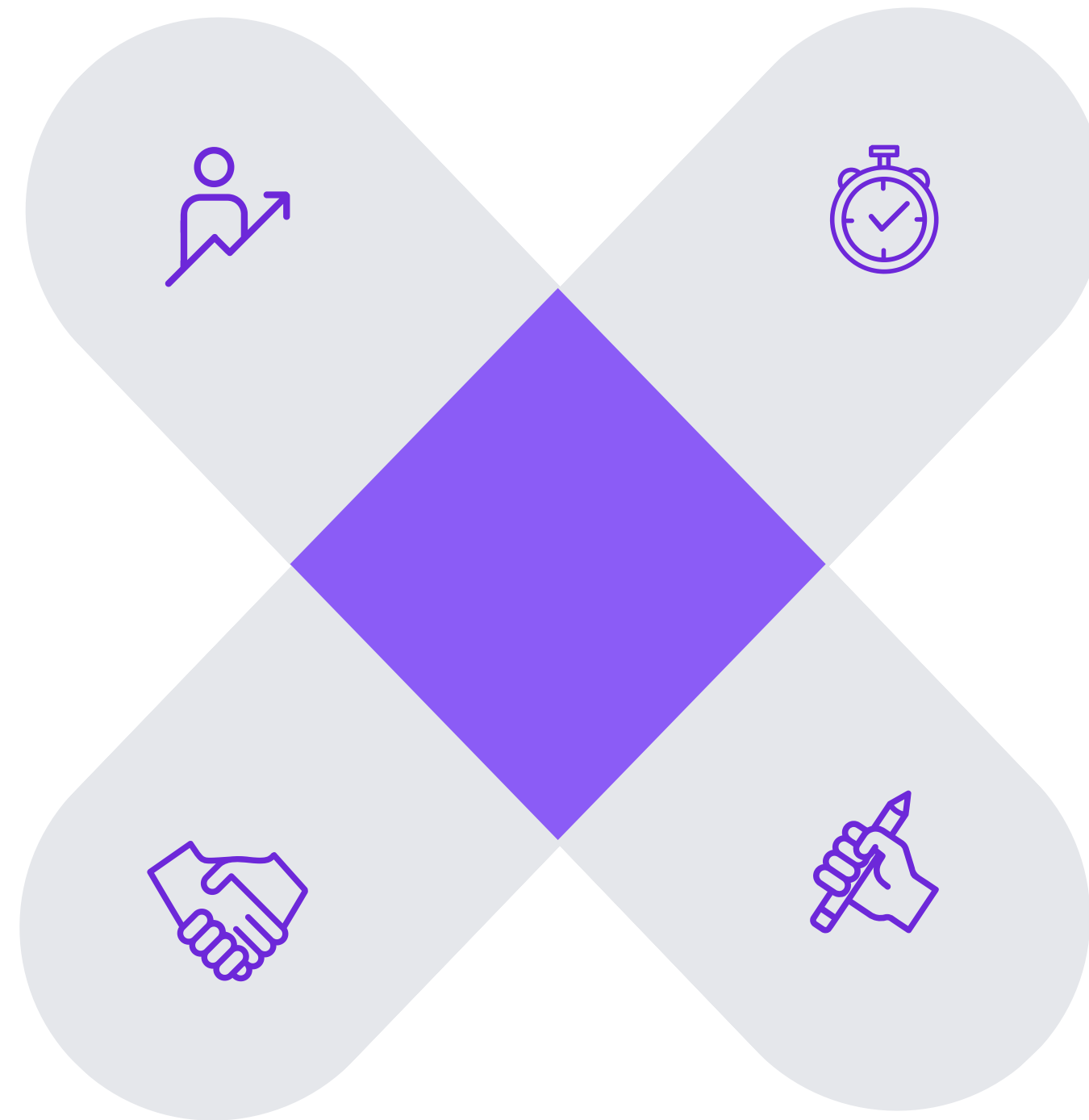
Our Research Question

The Problem

Drug interactions are difficult to study because biomedical knowledge is large and incomplete.

Our Question

What kind of biomedical information helps improve drug-drug interaction prediction?



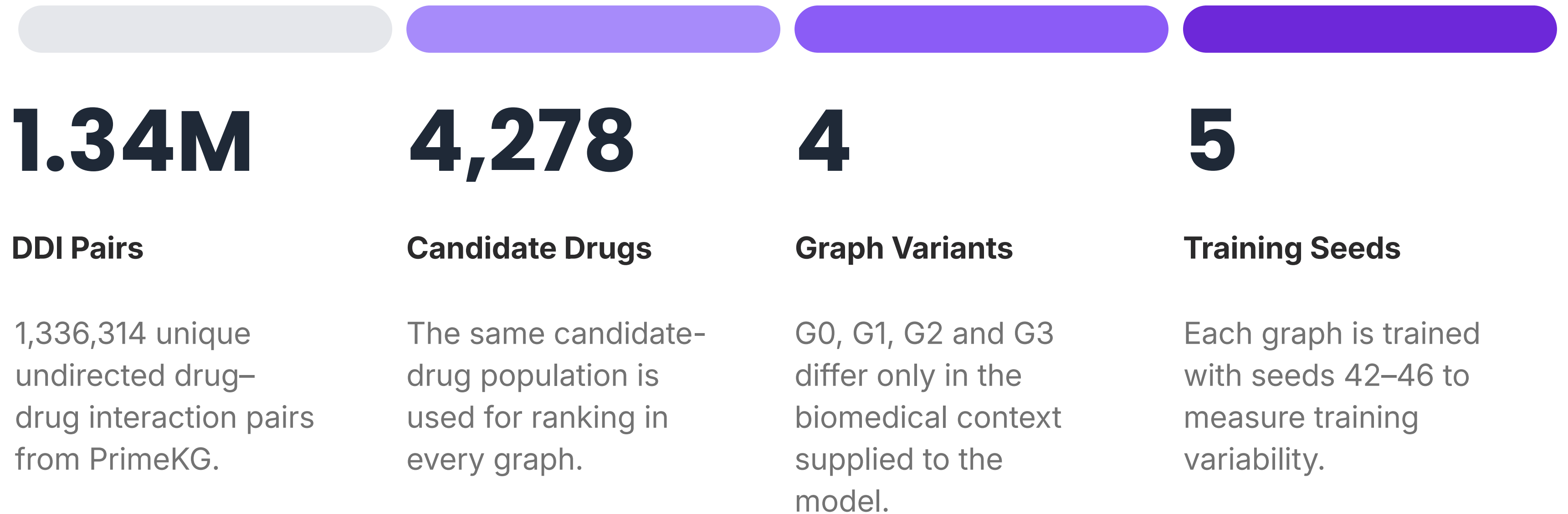
Our Approach

Compare four knowledge graphs while keeping the same model and evaluation setup.

Our Goal

Measure whether gene/protein and disease context help the AI rank possible DDI links.

Study Setup



Four Graph Variants

G0 — DDI Only

Only drug–drug
interaction edges.



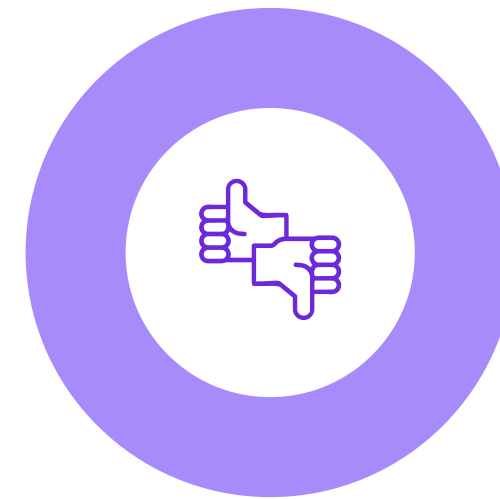
G1 — Gene / Protein

Adds Target,
Enzyme, Transporter
and Carrier relations.



G2 — Disease

Adds Indication,
Contraindication and
Off-label Use.



G3 — Combined

Combines both
gene/protein and
disease context.

How Does the Model Learn?



Knowledge Graph

Drugs and biomedical entities are connected by typed relationships.



Node Embeddings

Each node starts with a learned numerical representation.



R-GCN

The R-GCN learns from different kinds of neighboring relationships.



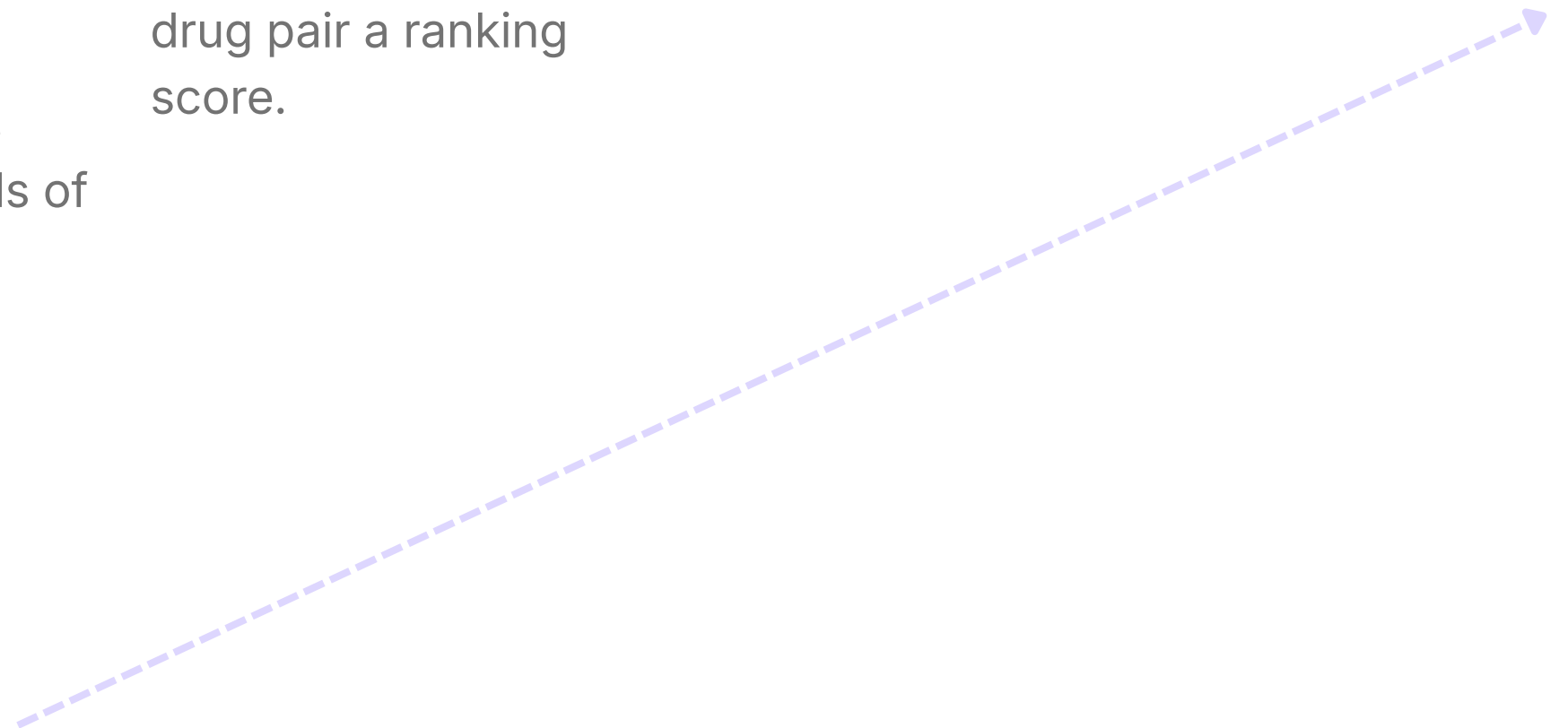
DDI Scoring

A symmetric decoder gives each possible drug pair a ranking score.



Ranking

The model ranks candidate drugs; higher-ranked true links are better.



How We Evaluate

Hits@1

How often the correct drug is ranked first.

Hits@5

How often the correct drug appears in the top 5.

Hits@10

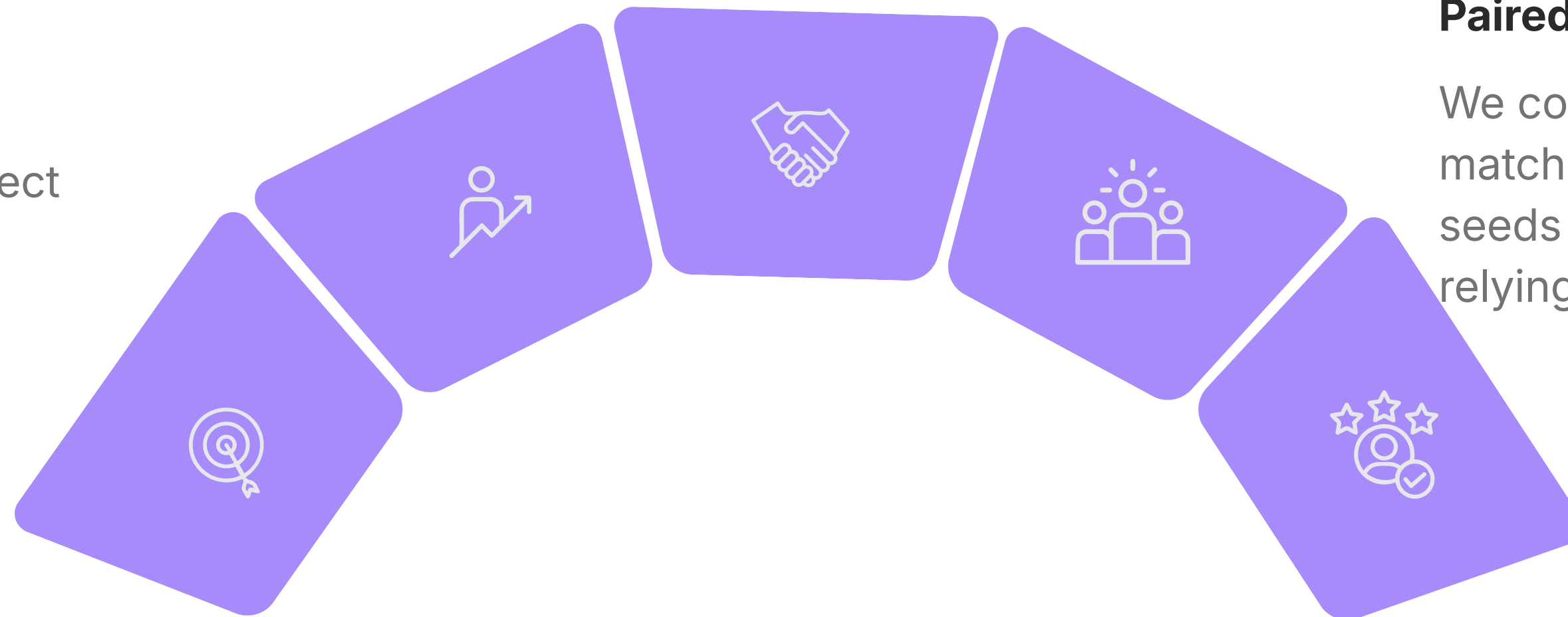
How often the correct drug appears in the top 10.

Paired Analysis

We compare matched training seeds instead of relying on one run.

MRR

How high the correct drug appears on average.



Main Results



G0 — DDI Only

Mean MRR: 0.527



G1 — + Gene / Protein

Mean MRR: 0.531 —
a small descriptive
improvement.



G2 — + Disease

Mean MRR: 0.527 —
disease context
alone did not
improve the baseline.



G3 — Combined

Mean MRR: 0.534 —
the highest average
result.

From Data to Demo

Evaluation

Compare ranking metrics and uncertainty across seeds.

CHEERS Web App

Export the verified model into an interactive research application.

Step 1

Step 2

Step 3

Step 4

Step 5

PrimeKG

Build a clean DDI split and biomedical support relations.

Graph Variants

Create G0–G3 while keeping the experiment controlled.

R-GCN Training

Train the same R-GCN architecture across all graph variants.

Which Information Helped Most?

Target

Largest average individual improvement in the five-seed ablation.

Carrier

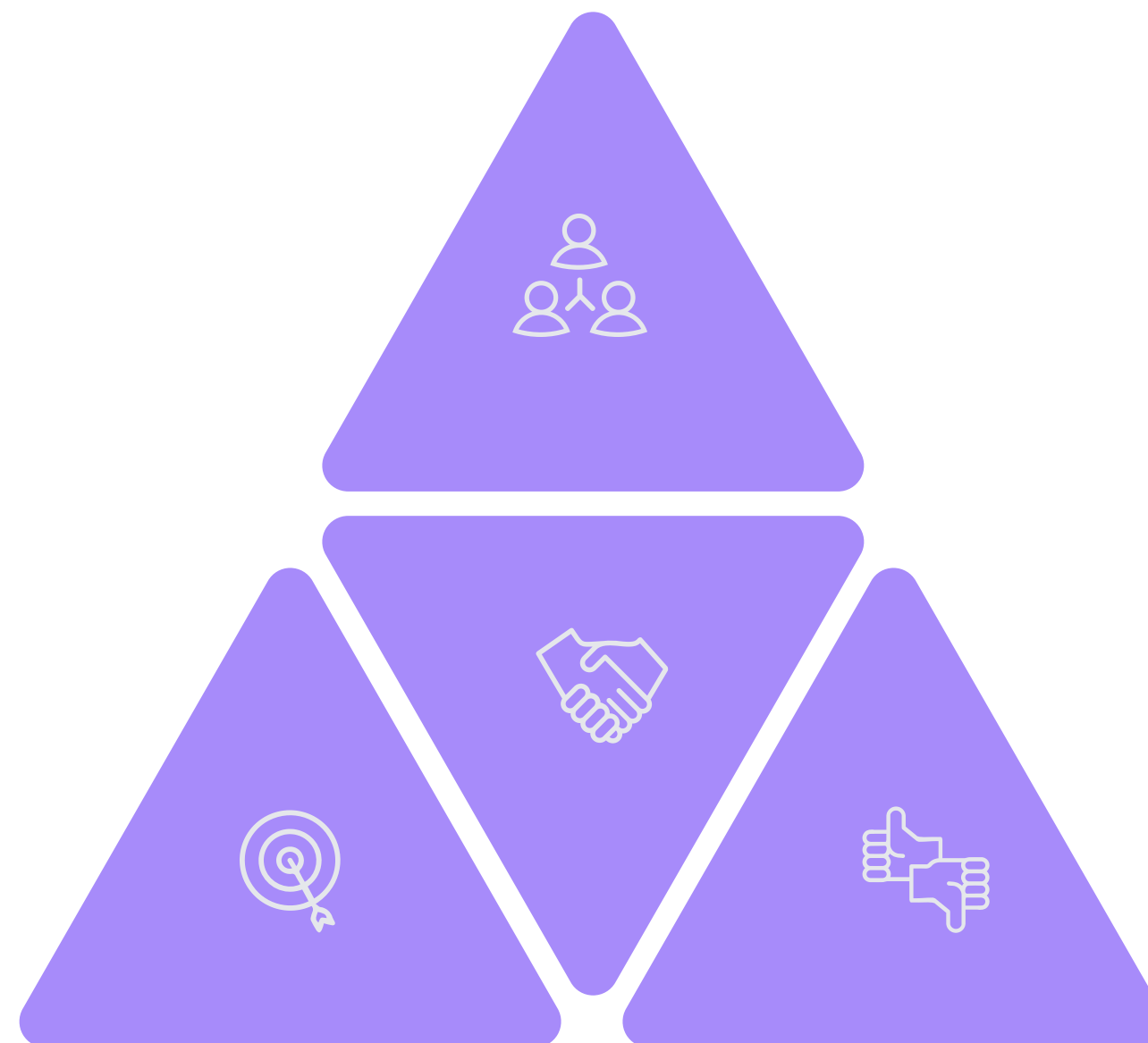
Small relation, but a positive average effect.

Transporter

Also showed a small positive average effect.

Uncertainty

No single relation was statistically established as better than G0.



CHEERS Web Application

DDI Predictor

Ranks possible unobserved drug-drug links for a selected drug.

Pair Graph Explorer

Shows shared gene/protein, disease and DDI graph context for a pair.



Subgraph Explorer

Explores one drug's neighborhood with relation and entity filters.

Evidence View

Retrieves FDA label information and PubMed literature separately from the model.

More System Features

Medicine Search

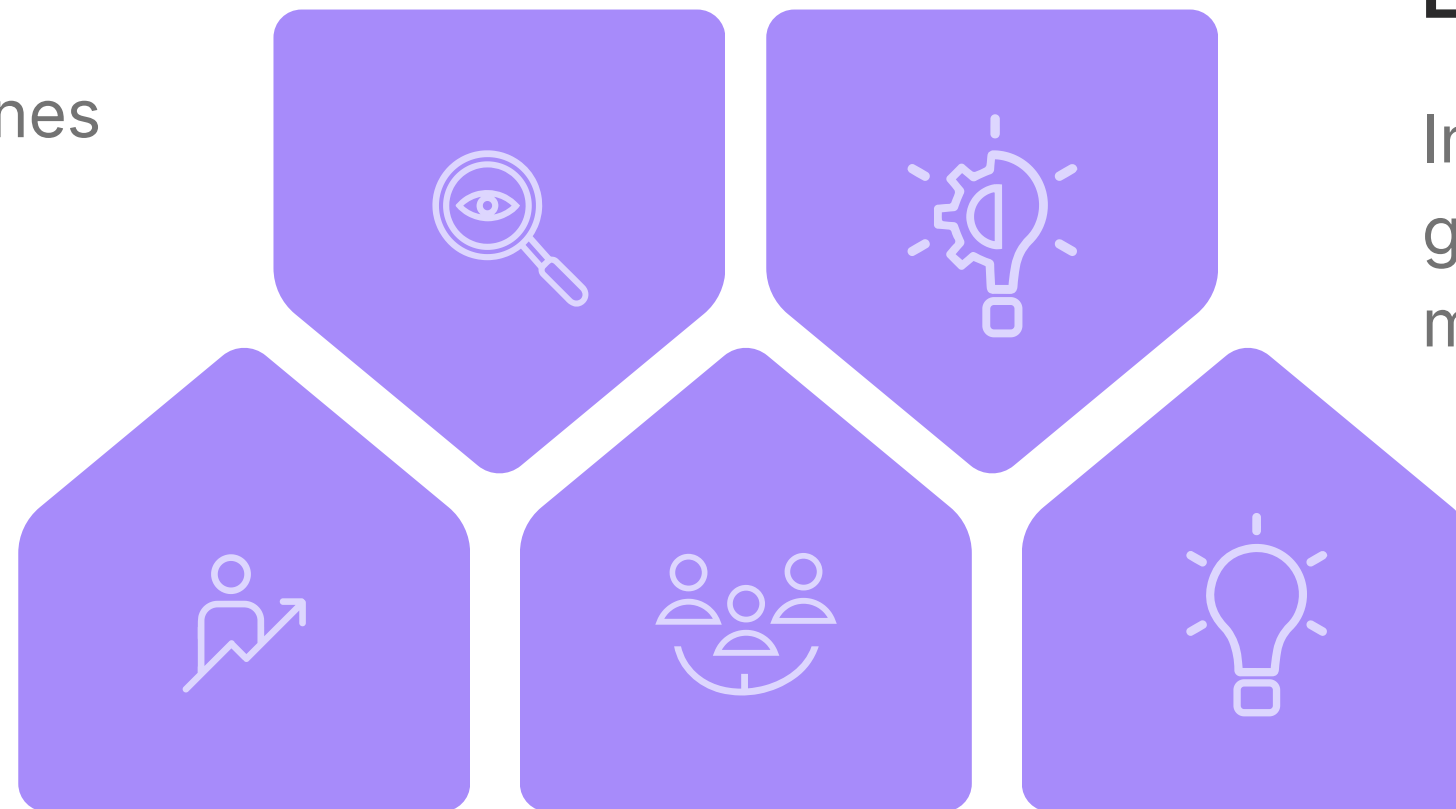
Search supported medicines and open research tools.

Entity Details

Inspect grounded drug, gene/protein and disease metadata.

PubMed

Retrieve related scientific literature for a selected drug pair.



FDA Labels

View relevant official label sections as independent evidence.

Medicine Label OCR

Read label text in the browser to help select a supported drug.

Key Takeaways

More ≠ Better

Adding more biomedical information does not automatically improve prediction.

No Single Winner

No individual relation was statistically established as the source of improvement.



Combined Context

G3 produced the highest average internal ranking performance.

Interactive System

CHEERS makes predictions, graph context and research results easier to explore.

Research Prototype

Model scores are ranking scores, not medical probabilities or treatment advice.

Thank You

A decorative graphic on the right side of the slide. It features a dark blue arc that starts from the top right and curves downwards. A purple arc starts from the bottom left, curves upwards, and crosses over the dark blue arc.

Team CHEERS

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Science and Engineering